

Embolization of the Middle Meningeal Artery for the Prevention of
Chronic Subdural Hematoma Recurrence in High Risk Patients
“EMPROTECT”

PROTOCOL FOR INTERVENTIONAL RESEARCH INVOLVING THE HUMAN PERSON ON
A MEDICAL DEVICE OR ON AN IN VITRO DIAGNOSTIC MEDICAL DEVICE

Version N°1.0 of 10/10/2019

Project code: PAOR18104

Coordinating investigator :

Dr Eimad Shotar

Neuroradiology Department, Interventional Neuroradiology Unit

Pitié-Salpêtrière Hospital

Tel: +33 1 84 82 73 66

E-mail: eimad.shotar@aphp.fr

Scientific lead :

Prof. Frédéric Clarençon

Neuroradiology Department, Interventional Neuroradiology Unit

Pitié-Salpêtrière Hospital

Tel: + 33 1 42 16 35 43

E-mail: frederic.clarencon@aphp.fr

Sponsor :

AP-HP and by delegation: Department of Clinical Research and Innovation (DRCI)

Hôpital Saint-Louis, 1, avenue Claude Vellefaux

Structure responsible for monitoring the research :

Clinical Research Unit (URC)

Pitié-Salpêtrière Charles Foix Hospital Group

Methodologist

Dr Agnes Dechartres

Tel: +33 1 42 16 05 99

E-mail: agnes.dechartres@aphp.fr

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Each addendum and the register of addendum versions are attached, independently of the protocol.

Each addendum can be modified (change of addendum version) without modifying the protocol version.

19. LIST OF ADDENDA

19.1 Addendum I: List of Investigators

CENTER	NAME	FIRST NAME	INSTITUTION	DEPARTMENT	ADDRESS	POSTAL CODE	CITY	COMPLEMENT	CONTACT Telephone	CONTACT Mail
ILE DE FRANCE										
	SHOTAR	Eimad	Pitié-Salpêtrière Hospital	Interventional Neuroradiology	47-83, boulevard de l'hôpital	75651	Paris	Cedex 13	01 84 82 73 66	eimad.shotar@aphp.fr
	MATHON	Bertrand	Pitié-Salpêtrière (APHP)	Neurosurgery	47-83, boulevard de l'hôpital	75651	Paris	Cedex 13	06 09 87 10 06	bertrand.mathon@aphp.fr
	NAGARRA	Olivier	Sainte Anne	Interventional Neuroradiology	1rue Cabanis	75674	Paris	Cedex 14, France	01 45 65 85 74	o.naggara@ch-sainte-anne.fr
	ZANELLO	Marc	Sainte Anne	Neurosurgery	1rue Cabanis	75674	Paris	Cedex 14, France	01 45 65 82 58	m.zanello@ch-sainte-anne.fr
	HOUDART	Emmanuel	Lariboisière (APHP)	Interventional Neuroradiology	2 Rue Ambroise Paré	75010	Paris		01 49 95 61 35	emmanuel.houdart@aphp.fr
	BERNAT	Anne-Laure	Lariboisière (APHP)	Neurosurgery	2 Rue Ambroise Paré	75010	Paris		01 49 95 52 87	annelaure.bernat@aphp.fr
	GOUTAGNY	Stéphane	Beaujon (APHP)	Neurosurgery	100 Boulevard du Général Leclerc	92110	Clichy			stephane.goutagny@aphp.fr
	TUILIER	Titien	Henri-Mondor (APHP)	Interventional Neuroradiology	51 Avenue du Maréchal de Lattre de Tassigny	94010	Créteil			titien.tuilier@gmail.com
	BRESSON	Damien	Henri-Mondor (APHP)	Neurosurgery	51 Avenue du Maréchal de Lattre de Tassigny	94010	Créteil			damien.bresson@aphp.fr
	ESCALARD	Simon	Rothschild Foundation	Interventional Neuroradiology	29 Rue Manin	75019	Paris			sescalard@for.paris

CENTER	NAME	FIRST NAME	INSTITUTION	DEPARTMENT	ADDRESS	POSTAL CODE	CITY	COMPLEMENT	CONTACT Telephone	CONTACT Mail
	CHAUVET	Dorian	Rothschild Foundation	Neurosurgery	29 Rue Manin	75019	Paris			dchauvet@for.paris
	FOUET	Mathilde	Percy Army Training Hospital	Neurosurgery	1 Rue du Lieutenant Raoul Batany	92190	Clamart			mathilde.fouet@intradef.gouv.fr
PROVINCE										
	MOUNAYER	Charbel	Limoges University Hospital	Interventional Neuroradiology	2, avenue Martin-Luther King	87042	Limoges		05 55 05 80 11	charbel.mounayer@unilim.fr
	SALLE	Henri	Limoges University Hospital	Neurosurgery	2, avenue Martin-Luther King	87042	Limoges		05 55 05 65 21	henrisalle1@gmail.com
	BRUNEL	Hervé	Marseille University Hospital	Interventional Neuroradiology	264 Saint-Pierre Street	13385	Marseille	Cedex 5	04 13 42 90 51	herve.brunel@ap-hm.fr
	DUFOUR	Henri	Marseille University Hospital (La Timone)	Neurosurgery	264 Saint-Pierre Street	13385	Marseille	Cedex 5	04 91 38 55 43	henry.dufour@ap-hm.fr
	TROUDE	Lucas	Marseille University Hospital (Hopital Nord)	Neurosurgery	Chemin des Bourrelly,	13015	Marseille		04 91 96 86 20	Lucas.TROUDE@ap-hm.fr
	HERBRETEAU	Denis	CHU Tours	Interventional Neuroradiology	2 bis, boulevard Tonnellé	37044	Tours		02 34 37 96 15	denis.herbreteau@univ-tours.fr
	ZEMMOURA	Ilyess	CHU Tours	Neurosurgery	2 bis, boulevard Tonnellé	37044	Tours		02 18 37 08 13	ilyess.zemmoura@univ-tours.fr
	BRICOUT	Nicolas	Lille University Hospital (Roger Salengro Hospital)	Interventional Neuroradiology	Rue Emile Laine	59037	Lille	Cedex	03 20 44 64 68	nicolas1.bricout@chru-lille.fr
	LEJEUNE	Jean-Paul	Lille University Hospital	Neurosurgery	Rue Emile Laine	59037	Lille	Cedex	03 20 44 66 15	jean-paul.lejeune@chru-lille.fr

19.2 Addendum II: *List of pairs for each center*

Centers in Ile-de-France		
Center	Co-investigator Neurosurgeon	Co-investigator Interventional neuroradiologist
Pitié-Salpêtrière (APHP)	MATHON Bertrand	SHOTAR Eimad
Sainte Anne	ZANELLO Marc	NAGARRA Olivier
Lariboisière (APHP)	BERNAT Anne-Laure	HOUDART Emmanuel
Beaujon (APHP)	GOUTAGNY Stéphane	HOUDART Emmanuel (Hôpital Lariboisière)
Henri-Mondor (APHP)	BRESSON Damien	TUILIER Titien
Rothschild Foundation	CHAUVET Dorian	ESCALARD Simon
Percy Hospital	FOUET Mathilde	SHOTAR Eimad (Hôpital Pitié-Salpêtrière)
Centers outside Paris Region		
Center	Co-investigator Neurosurgeon	Co-investigator Interventional neuroradiologist
Limoges University Hospital	SALLE Henri	MOUNAYER Charbel
Marseille University Hospital (Hôpital de la Timone)	DUFOUR Henri	BRUNEL Hervé
Marseille University Hospital (Hôpital Nord)	ROCHE Pierre-Hugues	BRUNEL Hervé
Lille University Hospital	LEJEUNE Jean-Paul	BRICOUT Nicolas
Tours University Hospital	ZEMMOURA Ilyess	HERBRETEAU Denis

19.3 Addendum III: Form for reporting a Serious Adverse Event (SAE) occurring in the course of research involving the human body on a Medical Device (MD)

Medical Organization and University Relations Department (DOMU) Department of Clinical Research and Innovation (DRCI)	ASSISTANCE PUBLIQUE  HÔPITAUX DE PARIS Form for the notification of a Serious Adverse Event (SAE) occurring during research involving the human body on a Medical Device (MD)	RESERVED FOR THE SPONSOR VIGILANCE REFERENCE : EDM reference: REC-DTYP-0270
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As soon as the investigator becomes aware of the SAE, this form must be duly completed (3 pages), signed and returned without delay to DRCI's Vigilance department by e-mail (eig-vigilance.drc@aphp.fr).

In order to avoid duplication of reports, you can fax the Vigilance Sector notification forms to +33 (0)1 44 84 17 99 only in the event of an unsuccessful attempt to send them by e-mail.

Initial notification of an SAE ☐ Follow-up ☐ Follow-up N° | | | |

1. Identification of the research	
Acronym: EMPROTECT	Notification date : 2 0 dd mm yyyy
Search Code: PAOR18104 Risk: C	Date on which the investigator became aware of the SAE : 2 0 dd mm yyyy
Full research title: Embolization of the Middle Meningeal Artery for the Prevention of Chronic Subdural Hematoma Recurrence in High Risk Patients	

2. Identification of the investigating center	
Center name:	Investigator (last name/first name):
City and postal code:	Tel: Fax:
Department:	

3. Identification and background of the research subject	
Person reference: - - - n°centre - n° selection order - initial - initial last name first name	Medical/surgical/family history relevant to case assessment (attach anonymized HRC if applicable) : Date of subdural hematoma surgery: 2 0 dd mm yyyy
Sex : ☐ M ☐ F Date of birth : dd mm yyyy Age : years	
Weight : kg Size : cm	
Date consent signature: 2 0 dd mm yyyy	
Randomization date: 2 0 dd mm yyyy	☐ Control group (usual care) ☐ Experimental group (middle meningeal artery embolization) Randomization no.: Treatment no. :

4. Description of experimental DM(s) (cross out box if experimental DM(s) not used)						
Common name / Trade name (if CE marked)	Model / Version (including software)	Serial no.	Batch no.	Indication for which the DM was used	Start and end dates of participant's DM use (dd/mm/yy)	If DM sterile: Sterilization date and best-before date (dd/mm/yy)
EmboSphere®					from to the	from to the
.....					from to the	from to the
If accessory(ies), consumable(s)						
Designation	Model	Serial no.	Batch no.	Manufacturer's name	Expiry date (dd/mm/yyyy)	

.....					
.....					

5. Procedure(s) for implementing experimental DM(s)	Date (dd/mm/yyyy)	Timeline	
		Before the SAE occurred	After the SAE has occurred
Conscious sedation ☐ Not applicable	 2 0 	☐	☐
Local anesthesia	 2 0 	☐	☐
Arterial catheterization	 2 0 	☐	☐
Selective angiography	 2 0 	☐	☐
MMA embolization proper	 2 0 	☐	☐
Heparin therapy ☐ Not applicable	 2 0 	☐	☐

6. Procedure(s) and act(s) added by research (e.g. biopsies, MRI, etc.) (cross out box if procedure(s) and act(s) not performed)	Completion date (dd/mm/yyyy)	Timeline	
		Before the SAE occurred	After the SAE has occurred
CT angiography of the supra-aortic trunks	 2 0 	☐	☐
.....	 2 0 	☐	☐

7. Medication(s) administered in the course of research: for conscious sedation if necessary, local anaesthesia, heparin therapy if necessary, selective angiography and CT angiography of the supra-aortic trunks (contrast media).

Please list the drugs used in the research, specifying the name of the speciality in each case, rather than the International Non-proprietary Name (INN).

Specialty name	Track ¹⁾	Dosage (specify unit ex : mg/d)	Dates administration (from dd/mm/yy to dd/mm/yy)	Visit cours e ²⁾	Indication	Action taken 0: continuation without change in dosage 1: stop 2: decrease dosage 3: increase dosage 4: don't know	Causality of the SAE 0: not drug-related 1: drug-related 2: don't know
			from to the	☐			
			from to the	☐			
			from to the	☐			
			from to the	☐			
			from to the	☐			
			from to the	☐			

8. Concomitant medication(s) at the time of the SAE, excluding those used to treat the adverse event (complete the table below and, if necessary, the appendix relating to concomitant medications, or cross out the box if not applicable).

Acronym: EMPROTECT

Reference of the searcher: [] [] [] - [] [] [] - [] [] []
n°centre - n° selection order - initial - initial
last name first name

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VIGILANCE REFERENCE :

EDM reference : REC-DTYP-0270

Evolution of the event

☐ Deaths

- ☐ unrelated to the SAE
☐ in relation to the SAE

Date: [] [] [] [] | 2 | 0 | [] []
dd mm yyyy

☐ Subject not yet restored, please specify :

- ☐ Stable ☐ Improvement ☐ Worsening

☐ Resolved:

- ☐ without sequelae
☐ with after-effects, please specify :

Date: [] [] [] [] | 2 | 0 | [] []
dd mm yyyy

☐ Evolution unknown

[] [] [] []
hh min

10. Other etiology(ies) considered

☐ No ☐ Yes If yes, please specify:

11. Additional examination(s) carried out

☐ No ☐ Yes If yes, specify date, nature and results: [attach anonymized reports].....

12. According to the investigator, the serious adverse event is (several boxes possible)

Related to research :

☐ Yes: ☐ to the research DM(s): which one(s)?

Which one: EmboSphere® ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

☐ the procedure(s) for implementing the DM(s): which procedure(s)?

Which one? Conscious sedation ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: Local anesthesia ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: Arterial catheterization ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one : Selective angiography ... ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: MMA embolization ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: Heparin therapy ☐ Certain relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: ☐ Ceratin ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

☐ research procedure(s)/action(s): which one(s)?

Which one? CT angiography of the supra-aortic trunks ☐ Definite relationship ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

Which one: ☐ Relationship certain ☐ Probable relationship ☐ Possible relationship ☐ Unlikely relationship (not excluded)

☐ No: ☐ to the progression of the disease under investigation: chronic subdural hematoma

☐ concomitant drug(s) administered, which drug(s):

☐ an intercurrent illness, which:

☐ other, please specify:

19.4 Addendum IV: Incident or defect report form (Annex to the DM SAE notification form)

Medical Organization and University Relations Department (DOMU) Clinical Research and Innovation Delegation (DRCI)	 Incident or malfunction report form Appendix to the form for reporting a Serious Adverse Event (SAE) occurring during research involving the human body and relating to a Medical Device (MD)	RESERVED FOR THE SPONSOR VIGILANCE REFERENCE : EDM reference: REC-DTYP-0419
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As soon as the investigator becomes aware of an incident or defect, this report form must be duly completed, signed, attached to the corresponding SAE notification form and returned without delay to DRCI's Vigilance sector by e-mail (eig-vigilance.drc@aphp.fr).

Documents can only be faxed to the Vigilance Sector on +33 (0)1 44 84 17 99 if an unsuccessful attempt is made to send them by e-mail.

Acronym: EMPROTECT Initial notification of an SAE ☐

Follow-up ☐ Follow-up N° |__|__|

Search code: PAOR18104

Person reference: |__|__| - |__|__| - |__| - |__|

Center no. - selection order no. - initial - initial
last name first name

Date of notification of the SAE associated with this appendix: __/__/____ (dd/mm/yyyy)

1. MEDICAL DEVICE INVOLVED	
Common name :	Commercial name: EmboSphere®
Laboratory manufacturer :	CE marking :

Notifier	Investigator	Service stamp :
Name and function :	Name :	
Signature :	Signature :	

2. DEFECT* / INCIDENT	
Date of occurrence of defect/incident : __/__/____ (dd/mm/yyyy)	Delay between the date of occurrence of the defect/incident and the date of occurrence of the SAE:

Notifier	Investigator	Service stamp :
Name and function :	Name :	
Signature :	Signature :	

Time of occurrence: _ _ hh _ _ min ☐ Missing data	_ _ / _ _ / _ _ dd / hh / min
---	---------------------------------------

Nature of defect/incident (description of facts and circumstances) :

.....

.....

.....

.....

.....

.....

Conservatory measures :

☐ no, no precautionary measures have been taken

☐ yes, one (more) protective measure(s) has (have) been taken, specify which :

☐ description :date : __/__/____

☐ description :date : __/__/____

☐ description :date : __/__/____

3. DEFECT/INCIDENT REPORTING
Has the manufacturer been informed of the defect/incident?
☐ missing data
☐ no, no information has been sent to the manufacturer
☐ yes, information was sent to the manufacturer on __/__/____ (dd/mm/yyyy) (see attachment if available)

***Defect** (Regulation (EU) 2017/745 on medical devices): any defect in the identity, quality, durability, reliability, safety or performance of a device under investigation, including any malfunction, error of use or defect in the information provided by the manufacturer.

****Incident** (article R5212-15 of the French Public Health Code): a noxious and unintended reaction occurring during the use of a medical device in accordance with its intended purpose; a noxious and unintended reaction resulting from the use of a medical device that does not comply with the manufacturer's instructions; any malfunction or alteration in the characteristics or performance of a medical device; any erroneous indication, omission or inadequacy in the instruction leaflet, instructions for use or maintenance manual.

19.5 Addendum V: *Pregnancy notification and follow-up form*

Medical Organization and Relations Department Universities (DOMU) Department of Clinical Research and Innovation (DRCI)	 ASSISTANCE PUBLIQUE HÔPITAUX DE PARIS <i>Notification and follow-up of a pregnancy occurring during research on a medical device</i>	RESERVED FOR THE SPONSOR INTERNAL REFERENCE : EDM reference: REC-DTYP-0185
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**This form must be duly completed (2 pages), signed and returned without delay to DRCI's Vigilance Department.
by e-mail (eig-vigilance.drc@aphp.fr)**

Documents can only be faxed to the Vigilance Sector on +33 (0)1 44 84 17 99 if an unsuccessful attempt is made to send them by e-mail.

1. Identification of research Acronym: EMPROTECT Search code: PAOR18104		Initial notification ☐		Follow-up notification ☐		Follow-up no.	
		Notification date :					
		Date investigator became aware of pregnancy :					
Full Research Title: Embolization of the Middle Meningeal Artery for the Prevention of Chronic Subdural Hematoma Recurrence in High Risk Patients							
2. Identification of the investigating center							
Institution name: City and postal code: Department:				Investigator (last name/first name): Tel: Fax :			
3. Identification of the pregnant person							
Person reference: - - - n°centre - n° selection order - initial - initial last name first name Date of birth: Date of inclusion: Randomization date: Randomization group : ☐ control ☐ experimental Date of last menstrual period: And/or date of start of pregnancy:				Special case of paternal exposure : ☐ Yes ☐ No Person reference: - - - n°centre - n° selection order - initial - initial last name first name Date of birth: Date of inclusion: Randomization date: Randomization group : ☐ control ☐ experimental			
Exposure during pregnancy : Tobacco : ☐ no ☐ yes (specify number of packages/year) : ☐ stop (specify date) : ☐ pursuit Alcohol : ☐ no ☐ yes (specify OH units) : ☐ stop (specify date) : ☐ pursuit Drugs: ☐ no ☐ yes (specify substance) : ☐ stop (specify date) : ☐ pursuit Other (specify) :							
4. Maternal history							
Medical :				Surgical :			
Obstetrical: Specify if miscarriage, ectopic pregnancy, termination of pregnancy (medical or voluntary), death <i>in utero</i> , congenital malformation, non-malformative congenital/neonatal pathology, ... (<i>number, date and nature/reason if applicable</i>).							

5. Description of experimental DM(s) (cross out box if experimental DM(s) not used)						
Common name / Trade name (if CE marked)	Model / Version (including software)	Serial no.	Batch no.	Indication for which the DM was used	Start and end dates of participant's DM use (dd/mm/yy)	If DM sterile : Sterilization date and best-before date (dd/mm/yy)
EmboSphere®					from [][] [][] [][] to the [][] [][] [][]	from [][] [][] to the [][] [][]

.....					from to the	from to the
If accessory(ies), consumable(s)						
Designation	Model	Serial no.	Batch no.	Manufacturer's name	Expiry date (dd/mm/yyyy)	
.....						
.....						

87. Procedure(s) for implementing experimental DM(s)	Date (dd/mm/yyyy)	Timeline	
		Before pregnancy	During pregnancy
Conscious sedation ☐ Not applicable		☐	☐
Local anesthesia		☐	☐
Arterial catheterization		☐	☐
Selective angiography		☐	☐
MMA embolization proper		☐	☐
Heparin therapy ☐ Not applicable		☐	☐

98. Procedure(s) and act(s) added by research (e.g. biopsies, MRI, etc.) (cross out box if procedure(s) and act(s) not performed)	Completion date (dd/mm/yyyy)	Timeline	
		Before pregnancy	During pregnancy
CT angiography of the supra-aortic trunks		☐	☐
.....		☐	☐

Acronym: EMPROTECT

RESERVED FOR THE SPONSOR INTERNAL REFERENCE : REC-DTYP-0270

Person reference: - - -
n°centre - n° selection order - initial - initial
last name first name

109. Medication(s) administered for research purposes: for conscious sedation if necessary, local anaesthesia, heparin therapy if necessary, selective angiography and CT angiography of the supra-aortic trunks (contrast media).							
Please list the drugs used in the research, specifying the name of the speciality in each case, rather than the International Non-proprietary Name (INN).							
Specialty name	Track ¹⁾	Dosage (specify unit ex : mg/d)	Dates administration (from dd/mm/yy to dd/mm/yy)	Visit course ²⁾	Indication	Action taken 0: continuation without change in dosage 1: stop 2: decrease dosage 3: increase dosage 4: don't know	Causality of the SAE 0: not drug-related 1: drug-related 2: don't know
			from to the 	☐			
			from to the 	☐			
			from to the 	☐			

			from _ _ _ _ _ _ _ _ _ to the _ _ _ _ _ _ _ _ _	☐			
			from _ _ _ _ _ _ _ _ _ to the _ _ _ _ _ _ _ _ _	☐			
			from _ _ _ _ _ _ _ _ _ to the _ _ _ _ _ _ _ _ _	☐			

110. Concomitant medication(s) administered as part of treatment

(Cf. appendix "List of concomitant medications" completed : ☐ Yes ☐ Not applicable)

Trade name (preferred) or International Nonproprietary Name	Date of first administration	Date of last administration Or in progress	Route of administration ⁽¹⁾	Dosage / 24h
	_ _ _ _ _ _ 2_ 0_ _ _	_ _ _ _ _ _ 2_ 0_ _ _ ☐ In progress		
	_ _ _ _ _ _ 2_ 0_ _ _	_ _ _ _ _ _ 2_ 0_ _ _ ☐ In progress		
	_ _ _ _ _ _ 2_ 0_ _ _	_ _ _ _ _ _ 2_ 0_ _ _ ☐ In progress		

(1) Route of administration: VO=oral; IM=intramuscular; IV=intravenous; SC=subcutaneous or other (please specify)

11. Pregnancy follow-up

☐ Ultrasound. Date(s) and results to be specified (attach anonymized CRs) :

☐ Other examinations. Date(s) and results to be specified (attach anonymized CRs) :

12. Pregnancy in progress ☐ (fax a new form completed at the end of the pregnancy to follow up on the initial notification)
or outcome of pregnancy ☐ (complete below)

Date: |_|_|_| |_|_|_| |2_|0_|_|_| Term: |_|_|_| SA |_|_|_| J

☐ Miscarriage

→ Anatomopathological examination available : ☐ No ☐ Yes, specify the result:

☐ Ectopic pregnancy

→ Anatomopathological examination available : ☐ No ☐ Yes, specify the result:

☐ Termination of pregnancy → Reason:

→ Anatomopathological examination available : ☐ No ☐ Yes, specify the result:

☐ Childbirth : ☐ Spontaneous ☐ Induced ☐ vaginal delivery ☐ Caesarean section

Multiple birth : ☐ No ☐ Yes, specify number :

Fetal suffering : ☐ No ☐ Yes, please specify:

Stillborn: ☐ No ☐ Yes, please specify :

Normal placenta : ☐ Yes ☐ No, please specify :

Amniotic fluid : ☐ Clear ☐ Other, specify :

Anaesthesia : ☐ General ☐ Peridural ☐ Spinal anesthesia ☐ None

13. Newborn (If multiple birth, complete parts 1, 2, 3, 9 and 10 of a new form and fax it to us)

Gender : ☐ Male ☐ Female

Weight : |_|_|_| grams Height : |_|_|_| cm Head circumference : |_|_|_| cm

APGAR: 1 minute: 5 minutes: 10 minutes: _____

Congenital malformation(s) : ☐ No ☐ Yes, please specify :

Non-malformative congenital/neonatal pathology(ies) : ☐ No ☐ Yes, please specify :

Did the newborn receive special care at birth? ☐ No ☐ Yes, please specify : ☐ Not applicable

Notifier	Investigator	Service stamp :
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Name and function : Signature :	Name : Signature :	
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19.6 CE mark and EmboSphere® data sheet



ATTESTATION / CERTIFICATE N° 8684 rev. 10

Délivrée à Paris le 28 novembre 2018

Issued in Paris on November 28th, 2018

ATTESTATION CE / EC CERTIFICATE

Examen CE de la Conception (du produit) / EC Design Examination (of the product)

ANNEXE II point 4 de la directive 93/42/CEE relative aux dispositifs médicaux

ANNEX II section 4 DIRECTIVE 93/42/EEC concerning medical devices

Fabricant / Manufacturer

BIOSPHERE MEDICAL SA

Parc des Nations - Paris Nord 2, 383 rue de la Belle Etoile

95700 ROISSY EN FRANCE FRANCE

Catégorie du(des) dispositif(s) / Device(s) category

Matériau d'embolisation artérielle

Material for arterial embolization

Identification du(des) dispositif(s) / Identification of device(s)

EMBOSPHERE - Microsphères en seringue pré-remplie sous blister

(code GMDN : 60938)

Voir addendum

EMBOSPHERE - Microspheres in pre-filled syringe under blister

(GMDN code : 60938)

Voir addendum

GMED atteste qu'à l'examen des résultats figurant dans le(s) rapport(s) référencé(s) P176659, le(s) produit(s) énuméré(s) ci-dessus est (sont) conforme(s) aux exigences de l'annexe I de la directive 93/42/CEE.

GMED certifies that, on the basis of the results contained in the file(s) referenced P176659, the product(s) compile(s) with the requirements of the directive 93/42/EEC, annex 1

Début de validité / Effective date : **November 28th, 2018 (included)**

Valable jusqu'au / Expiry date : **November 30th, 2019 (included)**

GMED_004-1-10-01-2018

GMED - 8684 rev. 10
Renouvelle le certificat 8684-9



Lionel DREUX
Certification Director

GMED • Société par Actions Simplifiée au capital de 300 000 € • Organisme Notifié/Notified Body n° 0459
Siège social : 1, rue Gaston Boissier - 75015 Paris • Tél : 01 40 43 37 00 • gmed.fr

Identification des dispositifs / Identification of devices

EMBOSPHERE - Microsphères en seringue pré-remplie sous blister
EMBOSPHERE - Microspheres in pre-filled syringe under blister

Taille Nominale Nominal size (µm)	Code Couleur (étiquetage) Color code (labeling)	Code de référence par volume d'Embosphere en seringue pré-remplie Reference code by volume of Embosphere in pre-filled syringe				Catégorie du dispositif Device Class
		1 mL	2mL	5 x 1mL Seringues Syringes	5 x 2mL Seringues Syringes	
40-120	Orange / Orange	S110GH	S120GH	S110GH	S120GH	III
50-100	Gris / Grey	S010GH	S020GH	S010GH	S020GH	
100-300	Jaune / Yellow	S210GH	S220GH	S210GH	S220GH	
300-500	Bleu / Blue	S410GH	S420GH	S410GH	S420GH	
500-700	Rouge / Red	S610GH	S620GH	S610GH	S620GH	
700-900	Vert / Green	S810GH	S820GH	S810GH	S820GH	
900-1200	Violet / Purple	S1010GH	S1020GH	S1010GH	S1020GH	

7 alinéas / 7 indented lines

GMED 0459



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Director Certification

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ADD - 720 DM 0701-31 rev 6 du 01/08/2018

 		Embosphere® microsphères pour embolisation en seringue pré-remplie		Version : 009 Page : 1 / 5 Date : 12/04/2019																																		
1. Renseignements administratifs concernant le fabricant																																						
1.1	Nom : BioSphere Medical, SA																																					
1.2	Adresse : Parc des Nations - Paris Nord 2 383 rue de la Belle Etoile 95700 Roissy en France (France) Adresse Postale : BP 54289 Roissy en France 95958 Roissy Charles de Gaulle Cedex		Tel: 33 (0)1 48 17 25 25 Fax : 33 (0)1 49 38 02 68																																			
1.3	Correspondant Matériovigilance : Madame Alix Fonlladosa Responsable Affaires Réglementaires		Tel: 33 (0)1 48 17 25 29 Fax : 33 (0)1 49 38 02 68 e-mail : alix.fonlladosa@merit.com																																			
2. Informations sur le dispositif ou équipement																																						
2.1	Dénomination commune : Particules d'embolisation																																					
2.2	Dénomination commerciale : Embosphere®																																					
2.3	Code nomenclature : GMDN 60 938 - Particules d'embolisation, non-bioabsorbable Code CLADIMED : Embolisation vasculaire C52A																																					
2.4	Code LPPR* (ex TIPS si applicable) : 3254468 (seringues de 1 ml) - 3268772 (seringues de 2 ml) * « liste des produits et prestations remboursables » prévue à l'article L 165-1 du code de la sécurité sociale.																																					
2.5	Classe du DM : III (selon la règle 8 point 2 et la règle 17 annexe IX de la directive) Directive de l'UE applicable : 93/42/CEE du 14 juin 1993 relative aux dispositifs médicaux Selon Annexe II point 3 et point 4 Norme : EN ISO 13485 ; 2016 Organisme notifié : GMED, numéro : CE 0459 Attestation CE de conformité n°: 9415 et 8684 Date de première mise sur le marché dans l'UE : Mars 2003																																					
2.6	Descriptif du dispositif (avec photo, schéma, dimensions, volume, ...) : Voir photo au point 9. Les microsphères Embosphere® sont biocompatibles, sphériques et non résorbables. Elles sont hydrophiles pour faciliter leur passage dans les micro-cathéters et provoquer l'occlusion totale de la lumière du vaisseau. Les microsphères sont réparties dans des seringues stériles, à usage unique, à raison de 1 ml ou 2 ml de microsphères respectivement dans 6 ou 7 ml de sérum physiologique, dans un conditionnement unitaire. Elles sont calibrées et disponibles en 7 gammes de tailles afin de permettre au médecin de faire son choix en fonction du diamètre du vaisseau à emboliser.																																					
2.7	Références Catalogue : <table border="1" data-bbox="295 1288 1251 1545"> <thead> <tr> <th rowspan="2">Taille nominale (µm)</th> <th rowspan="2">Code couleur (liseré des étiquettes)</th> <th colspan="2">Référence par volume d'Embosphere®</th> </tr> <tr> <th>1 ml</th> <th>2 ml</th> </tr> </thead> <tbody> <tr> <td>40 - 120 µm</td> <td>Orange</td> <td>S110GH</td> <td>S120GH</td> </tr> <tr> <td>50 - 100 µm</td> <td>Gris</td> <td>S010GH</td> <td>S020GH</td> </tr> <tr> <td>100 - 300 µm</td> <td>Jaune</td> <td>S210GH</td> <td>S220GH</td> </tr> <tr> <td>300 - 500 µm</td> <td>Bleu</td> <td>S410GH</td> <td>S420GH</td> </tr> <tr> <td>500 - 700 µm</td> <td>Rouge</td> <td>S610GH</td> <td>S620GH</td> </tr> <tr> <td>700 - 900 µm</td> <td>Vert</td> <td>S810GH</td> <td>S820GH</td> </tr> <tr> <td>900 - 1200 µm</td> <td>Violet</td> <td>S1010GH</td> <td>S1020GH</td> </tr> </tbody> </table> <u>UCD</u> (Unité de Commande) : unitaire ou boîte de 5 unités <u>CDT</u> (Multiple de l'UCD) : unitaire ou boîte de 5 unités <u>QML</u> (Quantité minimale de livraison) : unitaire ou boîte de 5 unités <u>Descriptif de la référence</u> : Microsphères d'embolisation en solution. <u>Conditionnement / emballages</u> : Seringue « réservoir » en Cyclic Olefin Polymer (COP) - Zeonex® de 20 ml, avec un piston en polycarbonate coloré (le code couleur des pistons est le même que celui des étiquettes, et est lié aux tailles des microsphères), à embout luer-lock Zeonex®, stérile, conditionnée dans un blister. Boîte carton unitaire ou boîte de 5 unités avec notice				Taille nominale (µm)	Code couleur (liseré des étiquettes)	Référence par volume d'Embosphere®		1 ml	2 ml	40 - 120 µm	Orange	S110GH	S120GH	50 - 100 µm	Gris	S010GH	S020GH	100 - 300 µm	Jaune	S210GH	S220GH	300 - 500 µm	Bleu	S410GH	S420GH	500 - 700 µm	Rouge	S610GH	S620GH	700 - 900 µm	Vert	S810GH	S820GH	900 - 1200 µm	Violet	S1010GH	S1020GH
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 	Embosphere® microsphères pour embolisation en seringue pré-remplie	Version : 009 Page : 2 / 5 Date : 12/04/2019
	d'instructions. Voir dimensions et photos au point 9. <u>Etiquetage</u> : Sur seringue, blister et boîte avec indications conformes à la réglementation en vigueur NF EN 1041 et NF EN ISO 15223-1. Modèles d'étiquettes : Voir photos au point 9. <u>Codes-barres</u> : type GS1. Voir image au point 9. Leur composition est la suivante : • (01) + 14 chiffres = code GTIN de l'article (code article international) • (17) + 6 chiffres = date maximum de validité (AAMMJJ) • (10) + 7 caractères = numéro de lot de fabrication	
2.8	Principaux éléments constitutifs du dispositif : Polymère acrylique Précisions complémentaires : ✓ Absence de latex ✓ Absence de phtalates (DHP) ✓ Présence de produit d'origine animale : Gélatine porcine ✓ Aucun dispositif, ni accessoire associés	
2.9	<u>Domaine - Indications</u> : Voir la notice d'instructions	
3.	<u>Procédé de stérilisation</u> :	
	<u>Mode de stérilisation du dispositif</u> : Stérilisation vapeur Par : BioSphere Medical La validation de la stérilisation a été effectuée suivant la norme NF EN ISO 17665	
4.	<u>Conditions de conservation et de stockage</u>	
	<u>Conditions normales de conservation & de stockage</u> : Dans leur seringue et emballage d'origine dans un endroit sec, frais à l'abri de la lumière. <u>Durée de la validité du produit</u> : 3 ans. La date de péremption est indiquée (en année, mois et jour) sur l'étiquetage du produit.	
5.	<u>Sécurité d'utilisation</u>	
5.1	<u>Sécurité technique</u> : Voir la notice d'instructions Les microsphères Embosphere® doivent être utilisées exclusivement par des médecins spécialisés ayant une formation spécifique des procédures d'embolisation vasculaire. La taille et la quantité de microsphères doivent être sélectionnées avec soin en fonction de la lésion à traiter et autres considérations cliniques pertinentes. Passage à l'IRM possible. Le produit n'est pas radio-opaque, mais peut être injecté avec des produits de contraste.	
5.2	<u>Sécurité biologique (s'il y a lieu)</u> : Voir la notice d'instructions Vérifier l'intégrité du protecteur de stérilité avant usage ; ne pas utiliser en cas d'endommagement de la seringue ou du luer lock. Produit stérile, à usage unique. Toute seringue entamée ne doit pas être réutilisée. Ne pas restériliser. Toute manipulation doit être réalisée dans des conditions propres à conserver la stérilité.	
6.	<u>Conseils d'utilisation</u>	
6.1	<u>Mode d'emploi</u> : Voir la notice d'instructions	
6.2	<u>Indications</u> : Voir la notice d'instructions	
6.3	<u>Précautions d'emploi</u> : Voir la notice d'instructions	
6.4	<u>Contre-Indications</u> : Voir la notice d'instructions	
7.	<u>Informations complémentaires sur le produit</u>	
	<u>Bibliographie, rapport d'essais cliniques, ou d'études pharmaco-économiques, amélioration du service rendu : recommandations particulières d'utilisation (restrictions de prise en charge, plateau technique, qualification de l'opérateur, etc) ...</u> : Voir la notice d'instructions <u>Contrôles effectués sur les matières premières</u> : ✓ Contrôle des matières premières de synthèse ✓ Contrôle des éléments de conditionnement	

	Embosphere® microsphères pour embolisation en seringue pré-remplie	Version : 009 Page : 3 / 5 Date : 12/04/2019
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Contrôles effectués sur les produits finis :

- ✓ Avant stérilisation : Contrôle microbiologique
- ✓ Après stérilisation :
 - Contrôles visuels physiques et microscopiques
 - Contrôles biologiques et microbiologiques

Volatils résultants de la destruction : CO₂, CO, H₂O, Méthane, Ammoniac, Ethane.

8. Liste des documents réglementaires et marketing associés au dispositif (s'il y a lieu)

- ✓ 1 : Attestation CE de conformité du système complet
- ✓ 2 : Attestation CE de conformité des dispositifs
- ✓ 3 : Notice d'instructions
- ✓ 4 : Brochure

9. Images

Photo du produit : Dimensions du blister : 67 x 28 x 218 mm



Photo de l'étiquette sur la seringue :

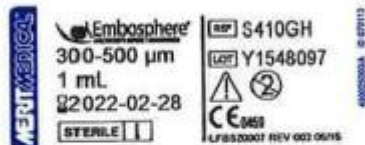


Photo de l'étiquette sur le tyvek :



Image du code-barres :



Dimensions de la boîte: 70 x 32 x 220 mm



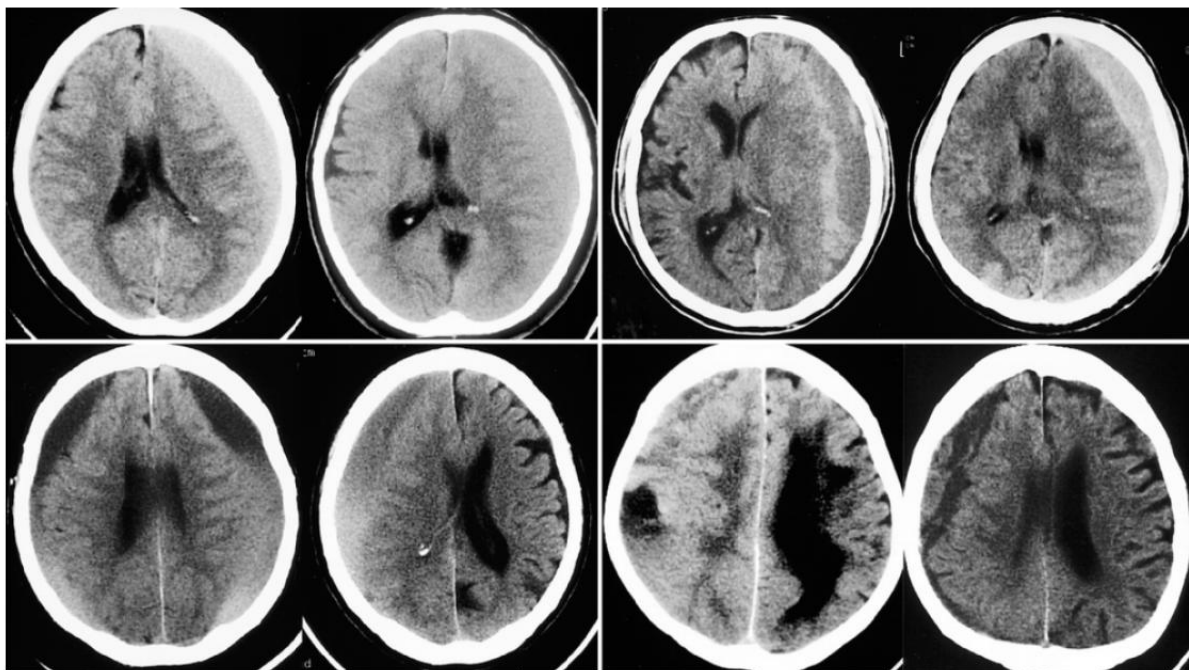
Etiquettes similaires pour les boîtes unitaires et les boîtes de 5 unités.

Photo de la boîte de 5 unités ; Dimensions de la boîte : 222 x 30 x 244 mm



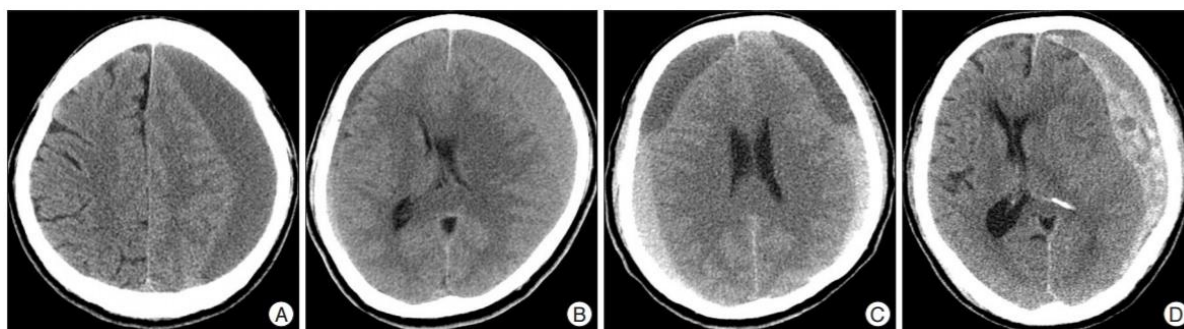
19.7 Addendum V: Radiological classification of cSDH

19.7.1 Radiological classification of cSDH according to Nakaguchi et al.



According to Nakaguchi et al. Factors in the natural history of chronic subdural hematomas that influence their postoperative recurrence. J Neurosurg. 2001;95:256-62. Upper Left: Homogeneous type. Upper Right: Laminar type. Lower Left: Separated. Lower Right: Trabecular type.

19.7.2 Radiological classification of HSDc according to Park et al.



According to Park H-R et al. Multiple Densities of the Chronic Subdural Hematoma in CT Scans. J. Korean Neurosurg. Soc. 2013;54:38-41. Chronic SDH classification according to CT density. A: Hypodensity (<25 HU). B: Homogeneous isodensity (25-35 HU). C: Layered type. D: Mixed type.